

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

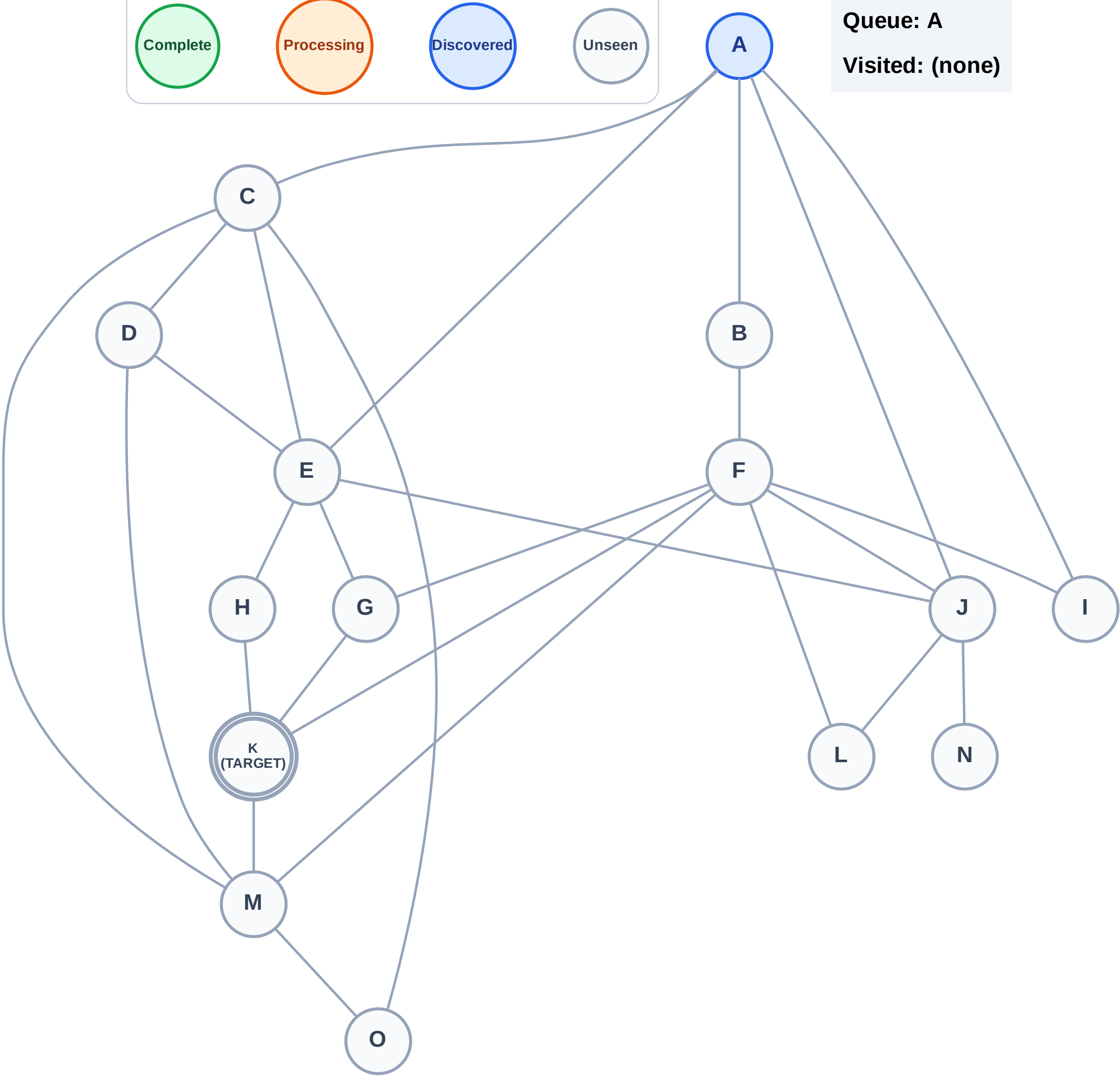
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

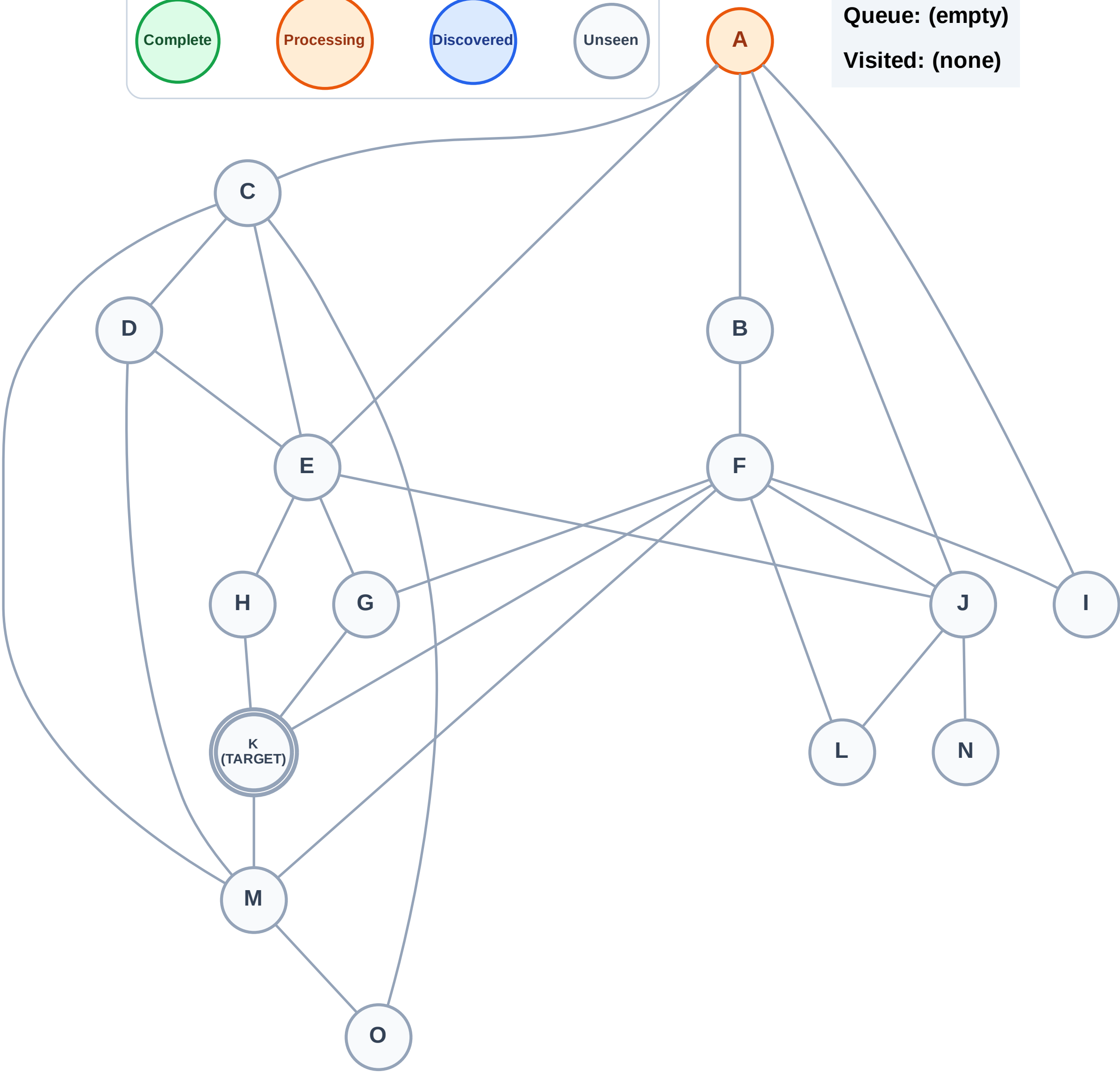
Step 2: Process node A

Legend



Queue: (empty)

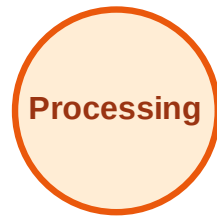
Visited: (none)



BFS Traversal

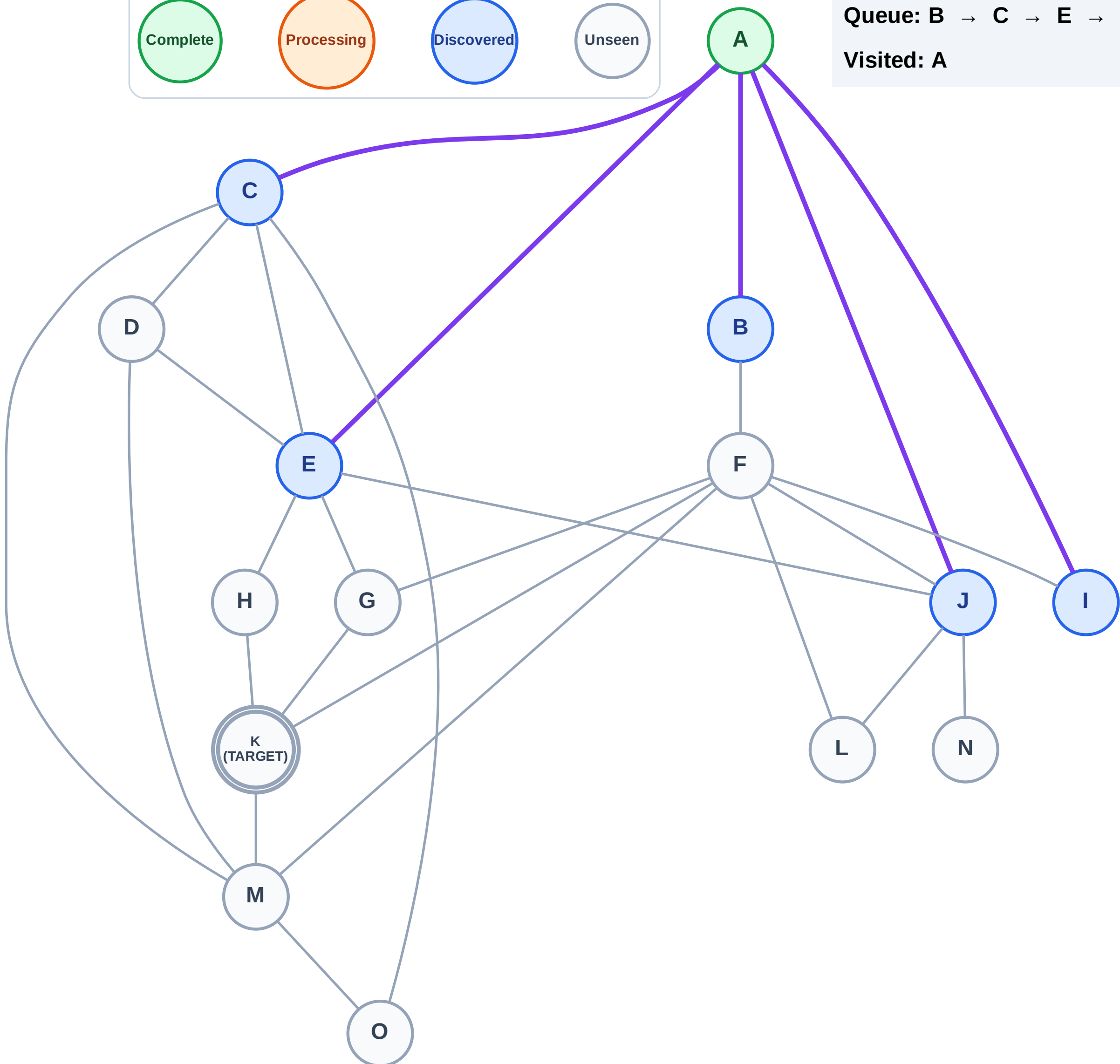
Step 3: Discover B, C, E, I, J from A

Legend



Queue: B → C → E → I → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

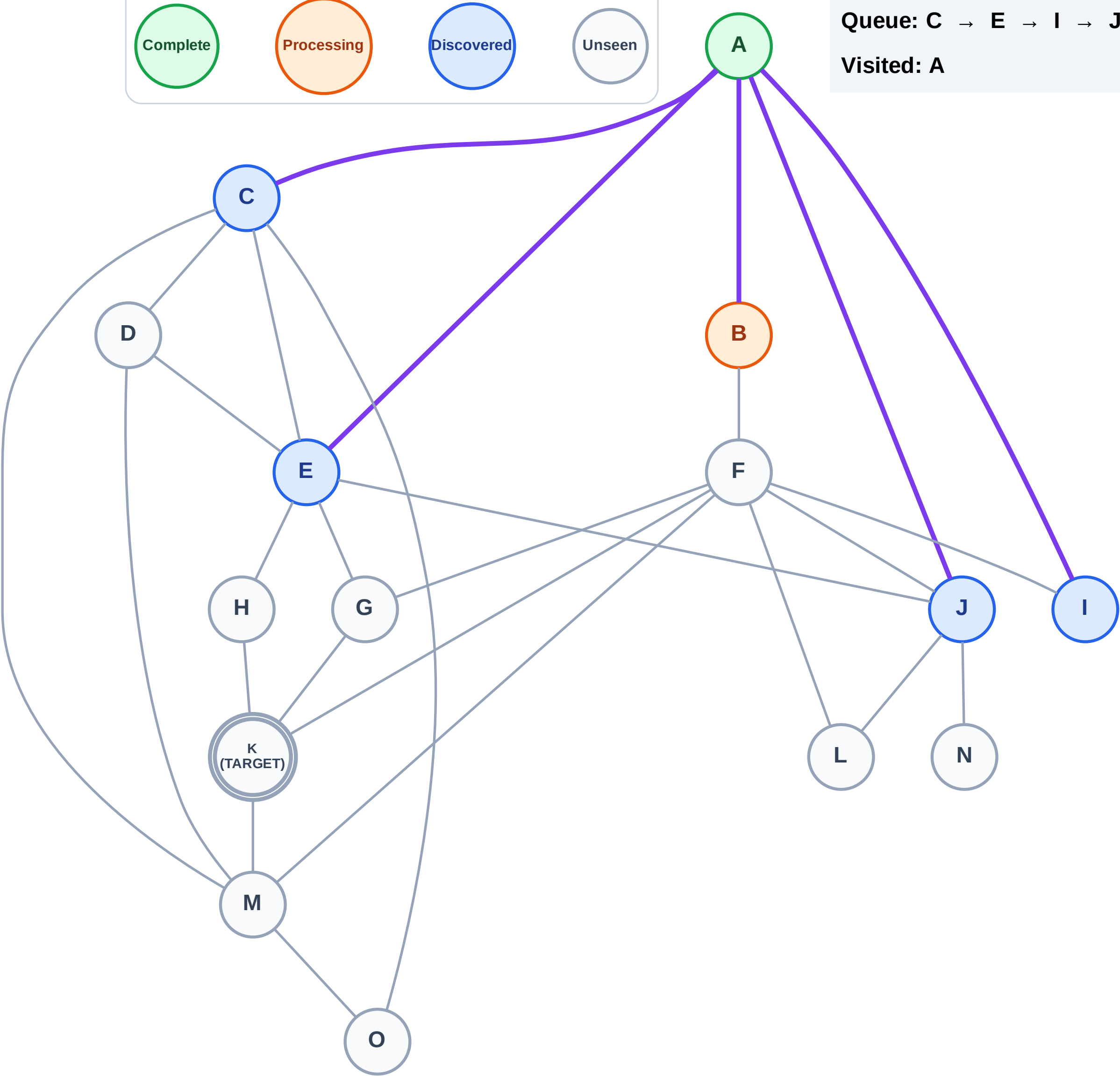
Processing

Discovered

Unseen

Queue: C → E → I → J

Visited: A



BFS Traversal

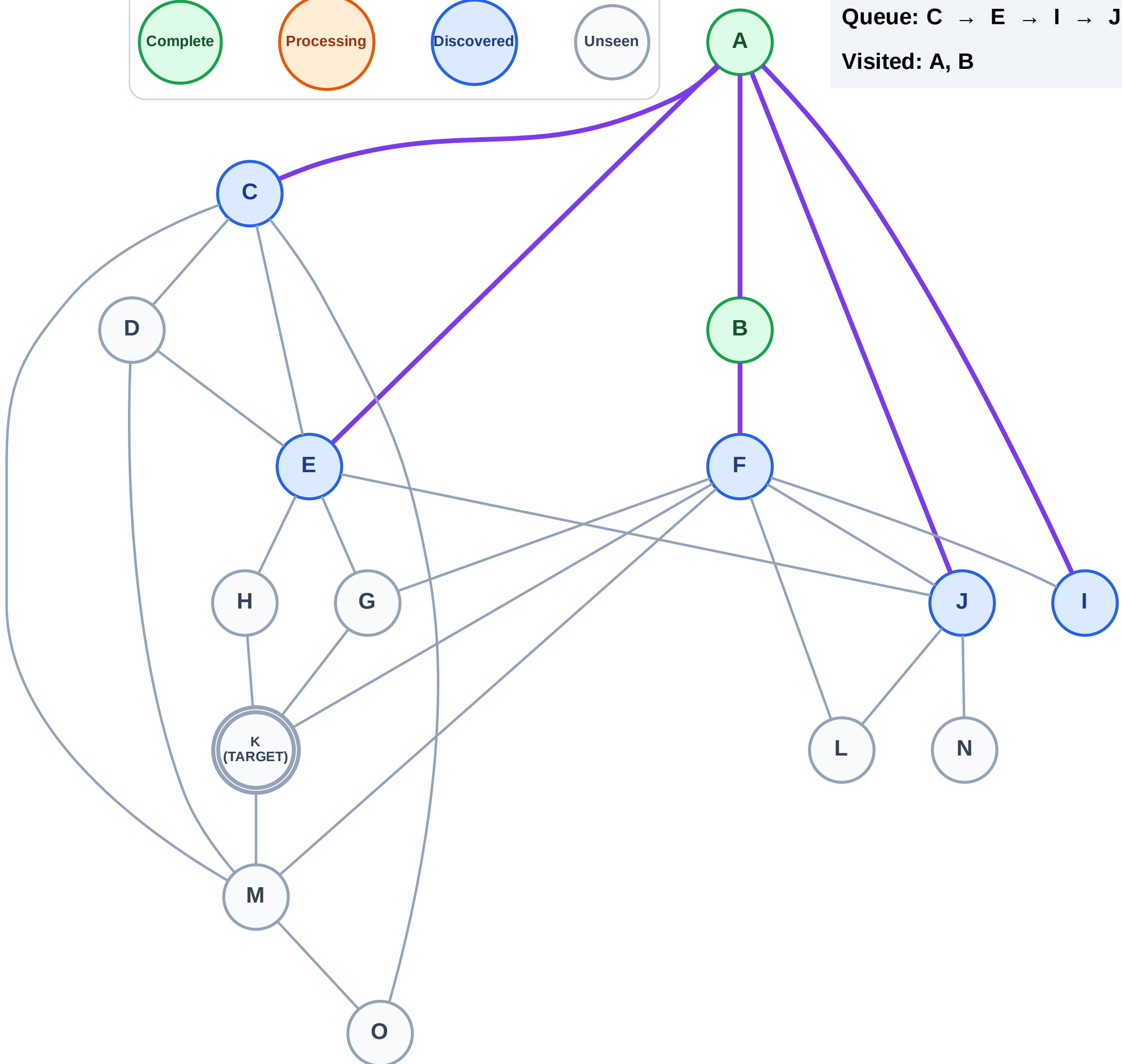
Step 5: Discover F from B

Legend



Queue: C → E → I → J → F

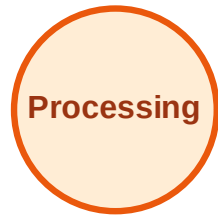
Visited: A, B



BFS Traversal

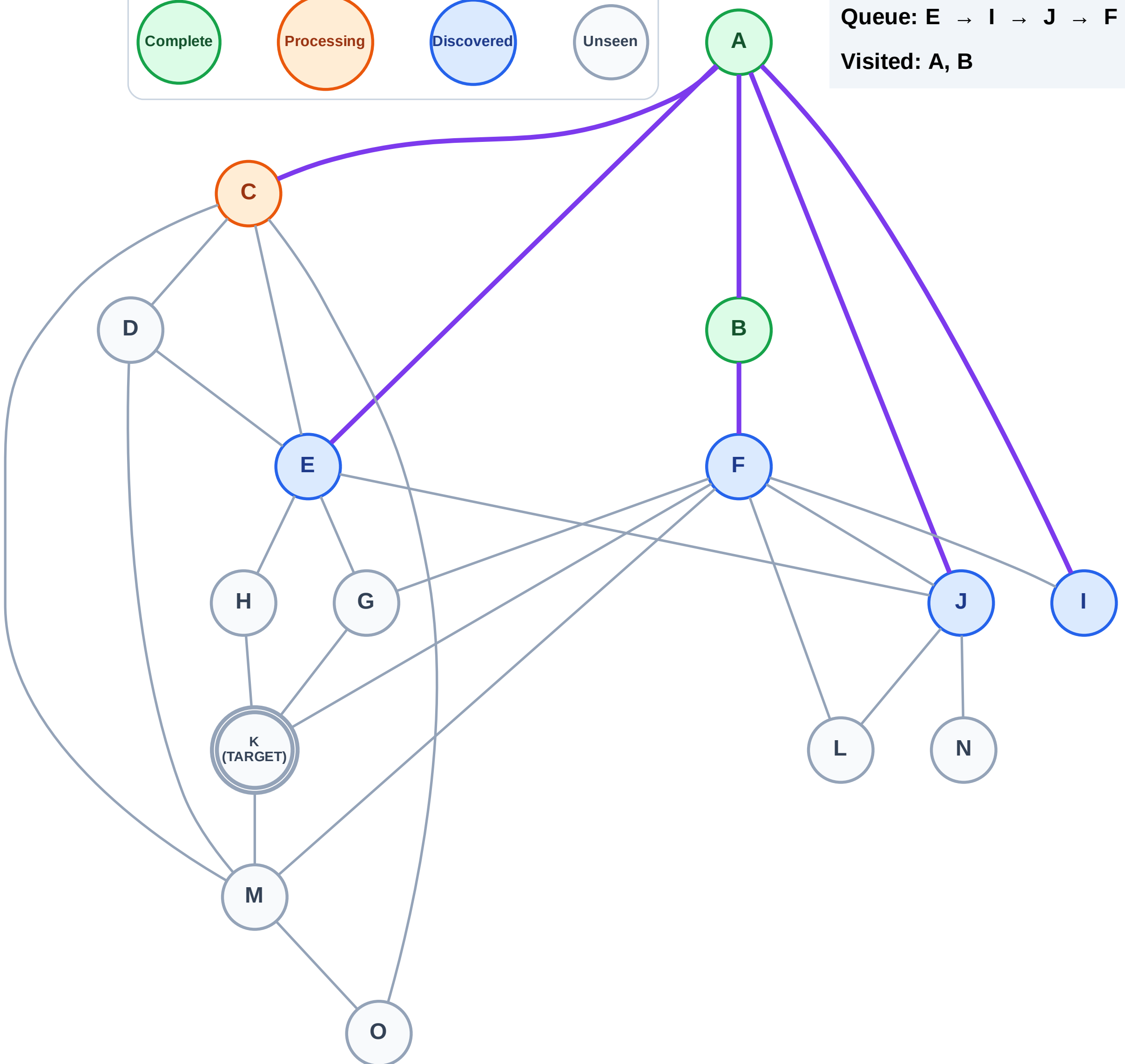
Step 6: Process node C

Legend



Queue: E → I → J → F

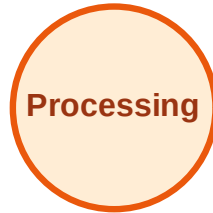
Visited: A, B



BFS Traversal

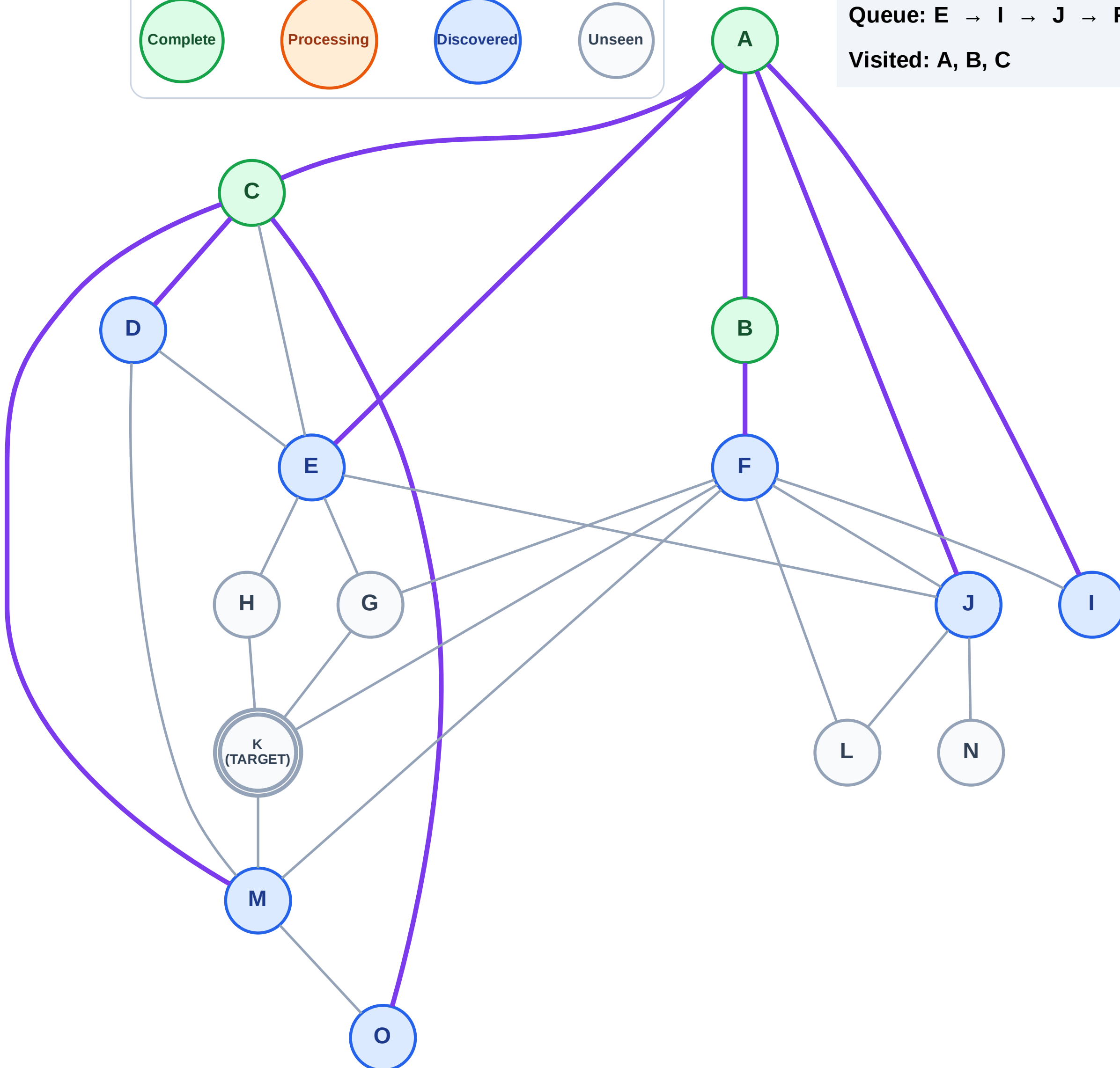
Step 7: Discover D, M, O from C

Legend



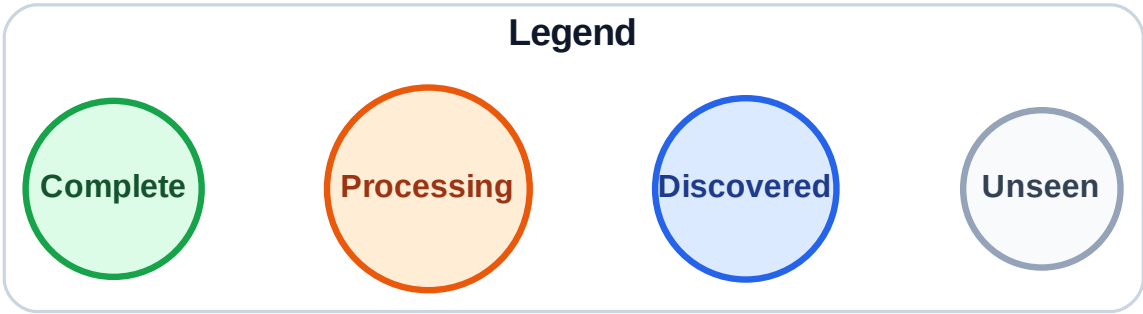
Queue: E → I → J → F → D → M → O

Visited: A, B, C



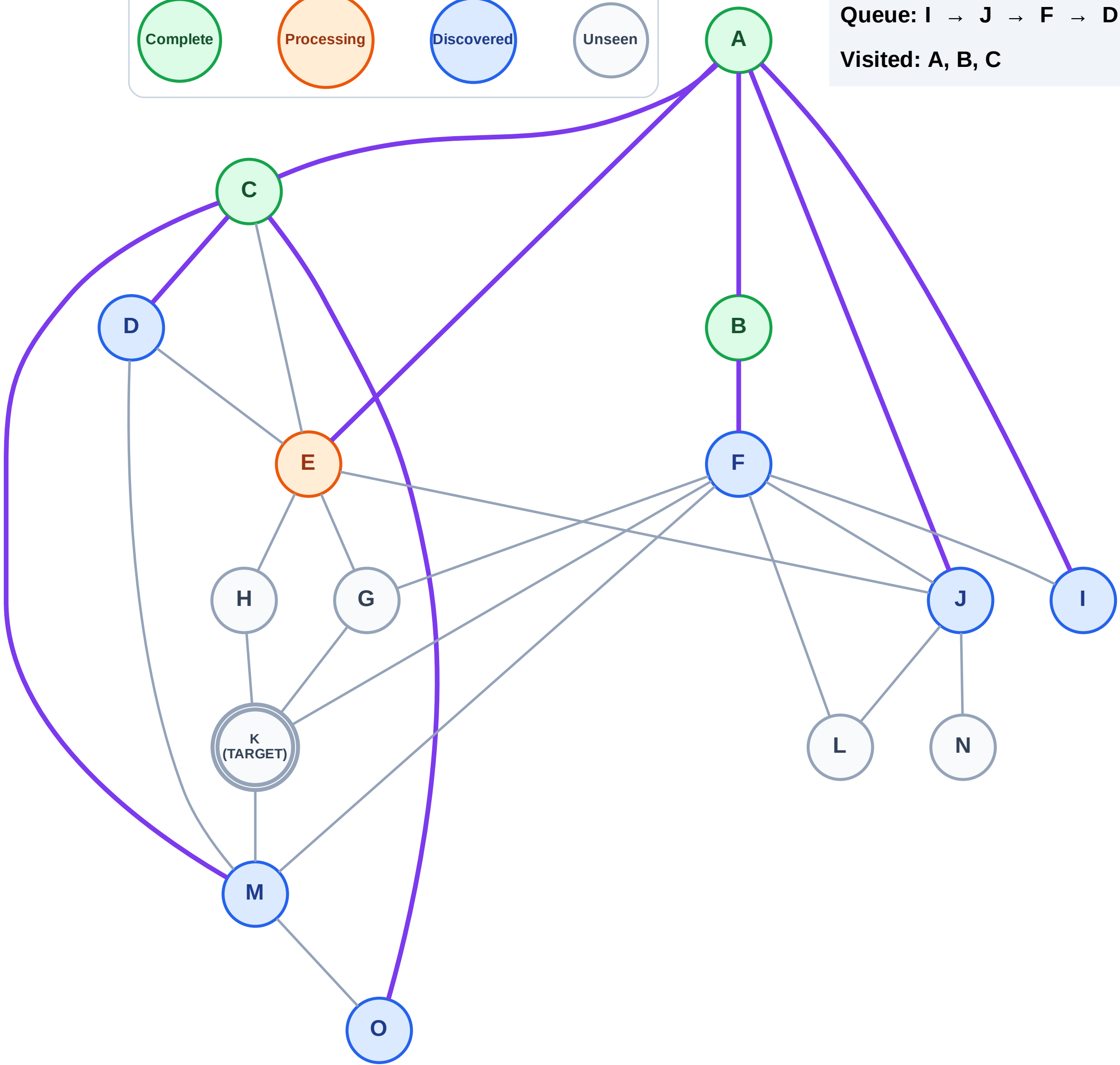
BFS Traversal

Step 8: Process node E



Queue: I → J → F → D → M → O

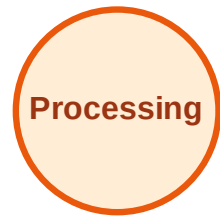
Visited: A, B, C



BFS Traversal

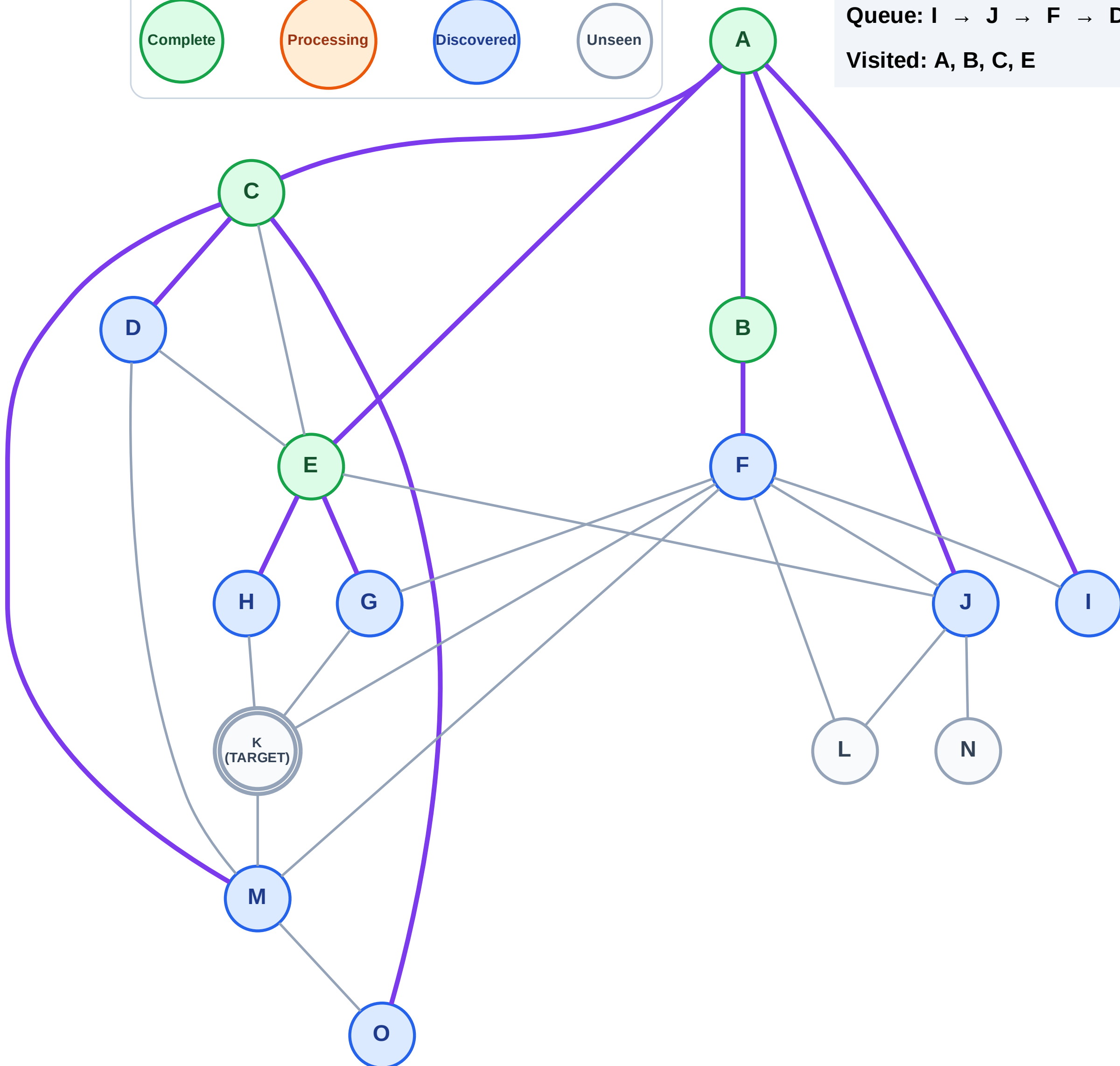
Step 9: Discover G, H from E

Legend



Queue: I → J → F → D → M → O → G → H

Visited: A, B, C, E



BFS Traversal

Step 10: Process node I

Legend

Complete

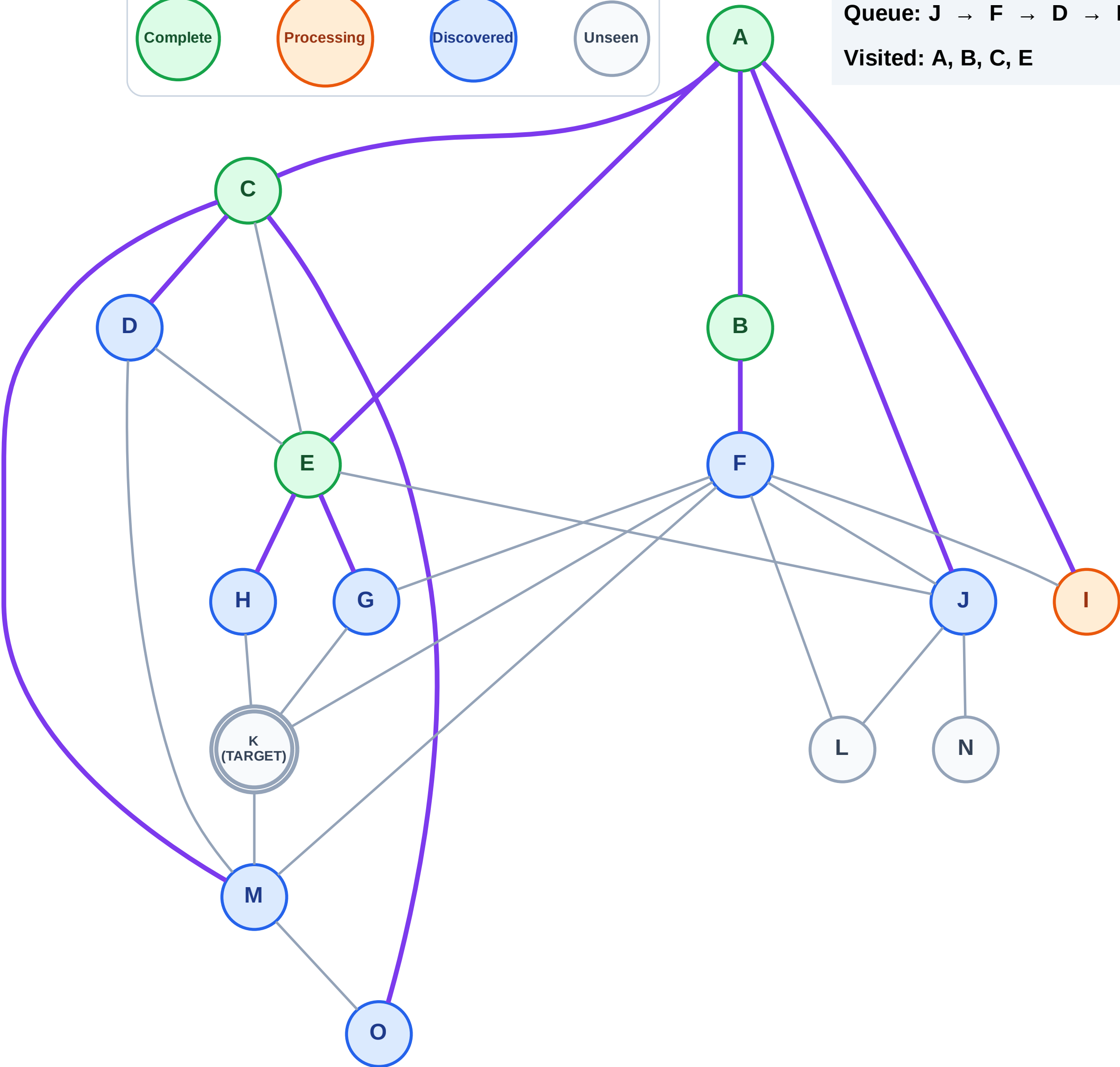
Processing

Discovered

Unseen

Queue: J → F → D → M → O → G → H

Visited: A, B, C, E



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

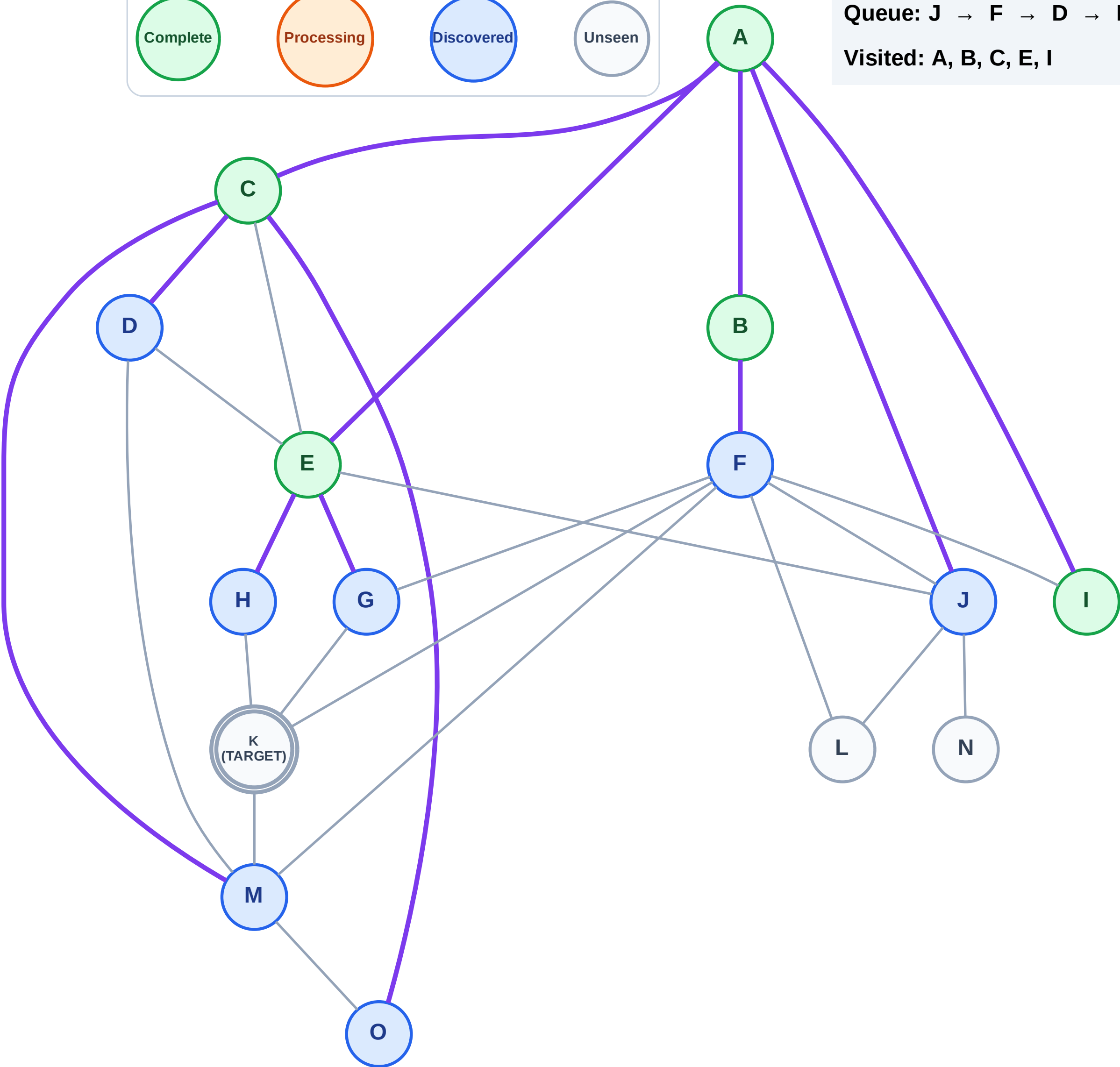
Processing

Discovered

Unseen

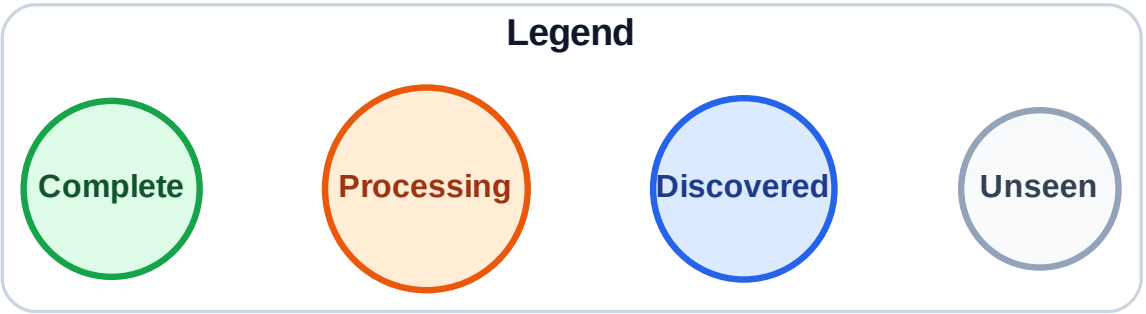
Queue: J → F → D → M → O → G → H

Visited: A, B, C, E, I



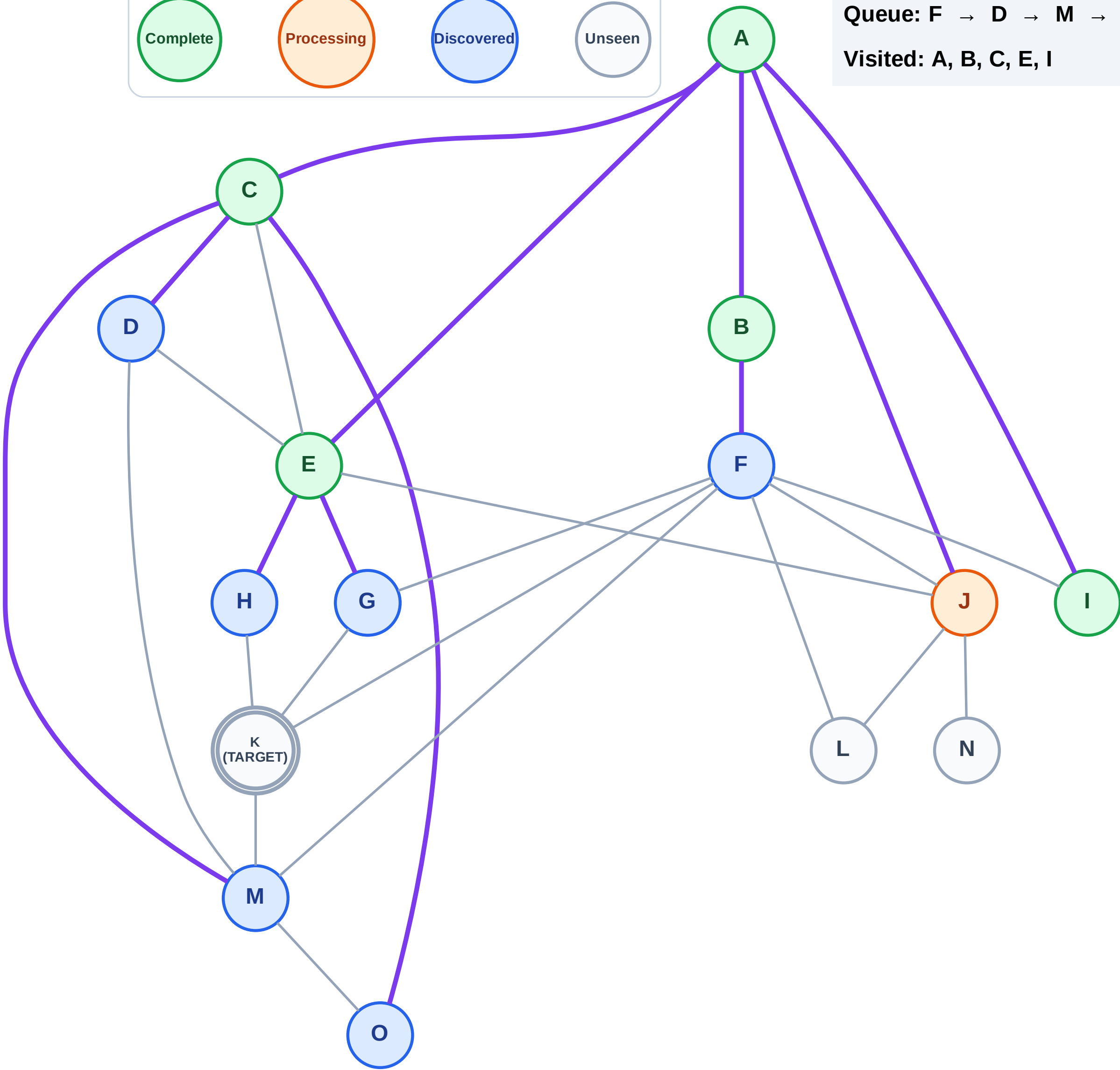
BFS Traversal

Step 12: Process node J



Queue: F → D → M → O → G → H

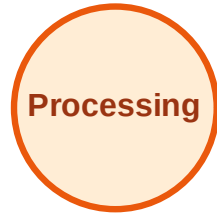
Visited: A, B, C, E, I



BFS Traversal

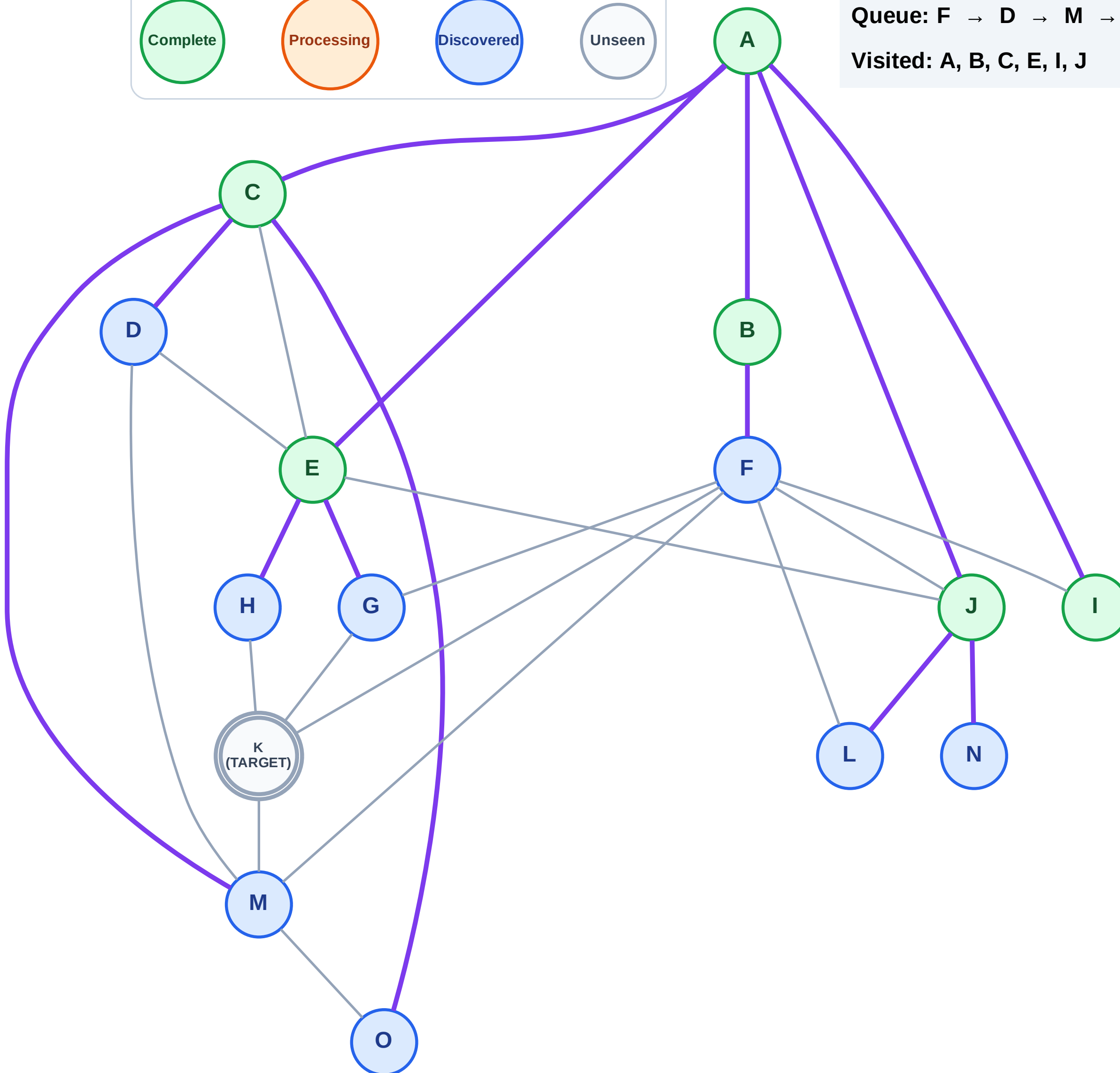
Step 13: Discover L, N from J

Legend



Queue: F → D → M → O → G → H → L → N

Visited: A, B, C, E, I, J



BFS Traversal

Step 14: Process node F

Legend

Complete

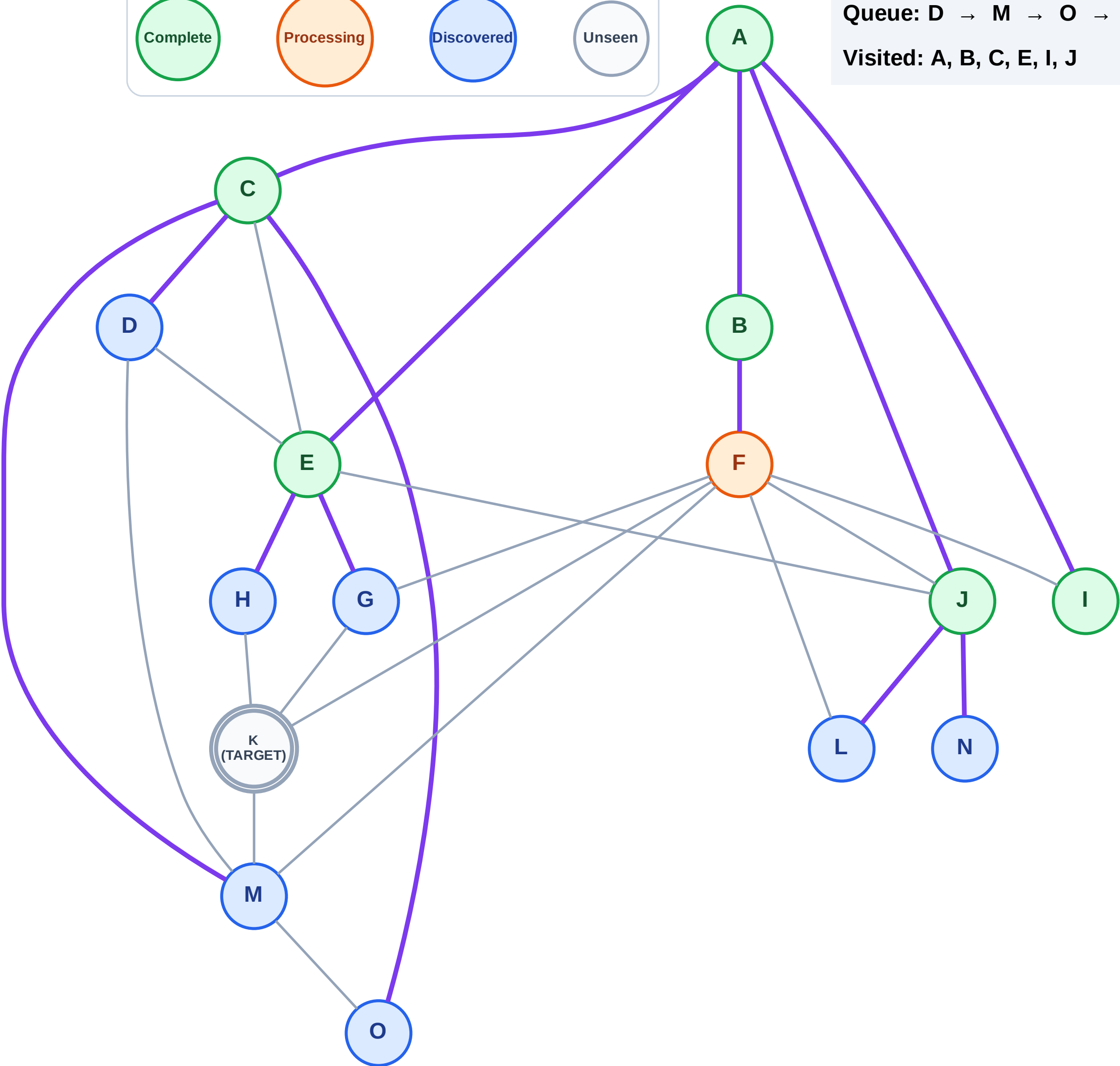
Processing

Discovered

Unseen

Queue: D → M → O → G → H → L → N

Visited: A, B, C, E, I, J



BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend

Complete

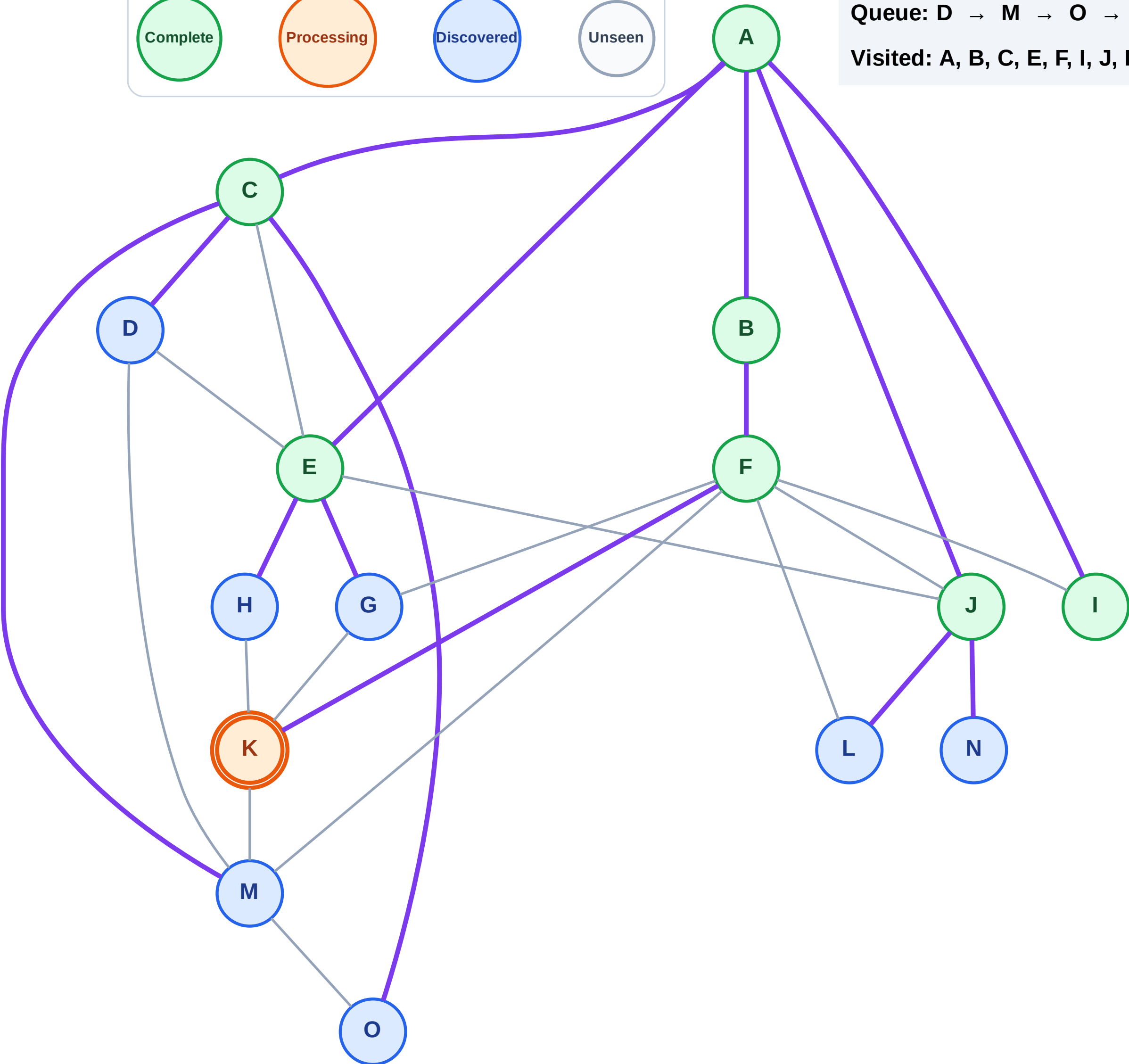
Processing

Discovered

Unseen

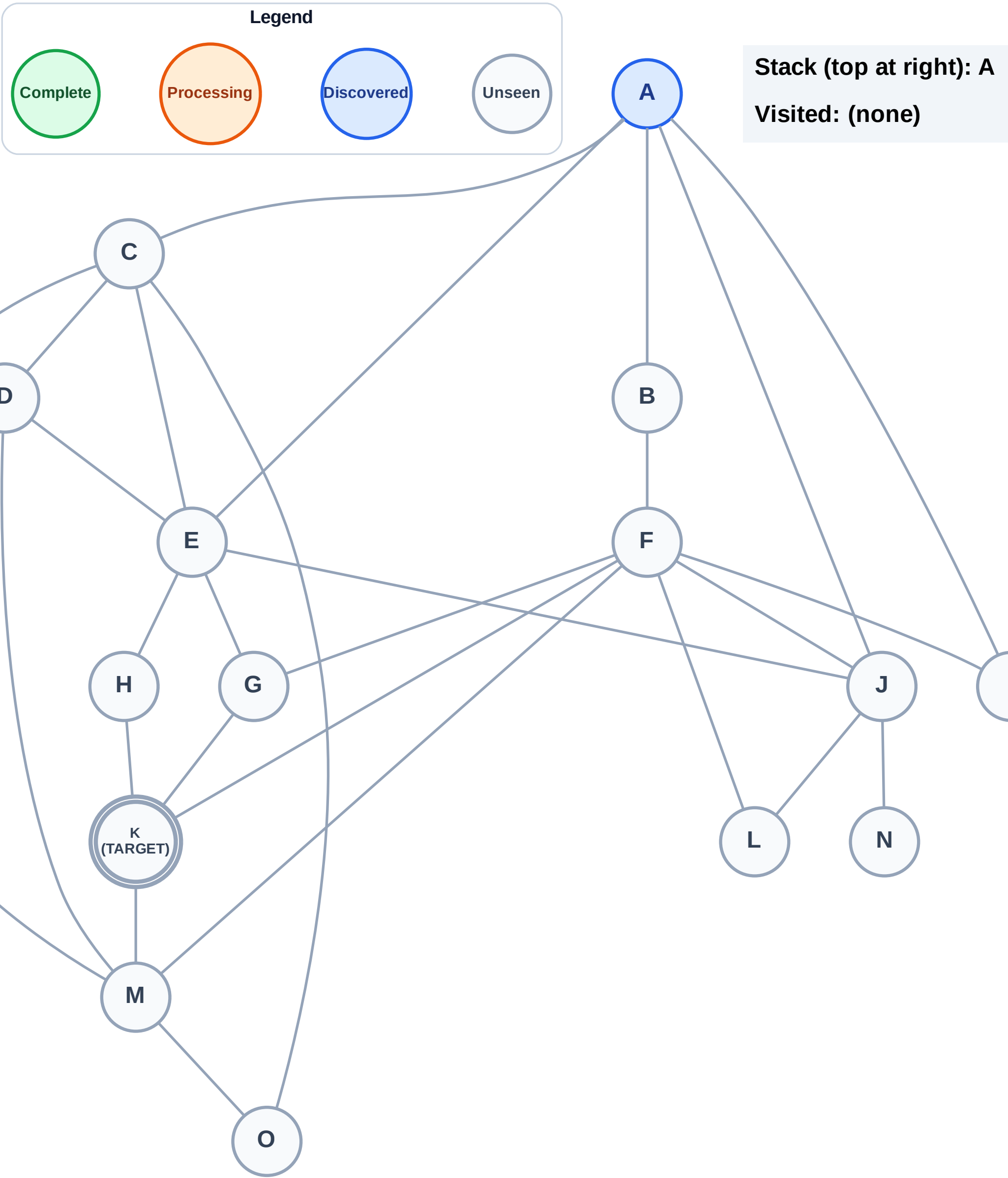
Queue: D → M → O → G → H → L → N

Visited: A, B, C, E, F, I, J, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



DFS Traversal

Step 2: Process node A

Legend

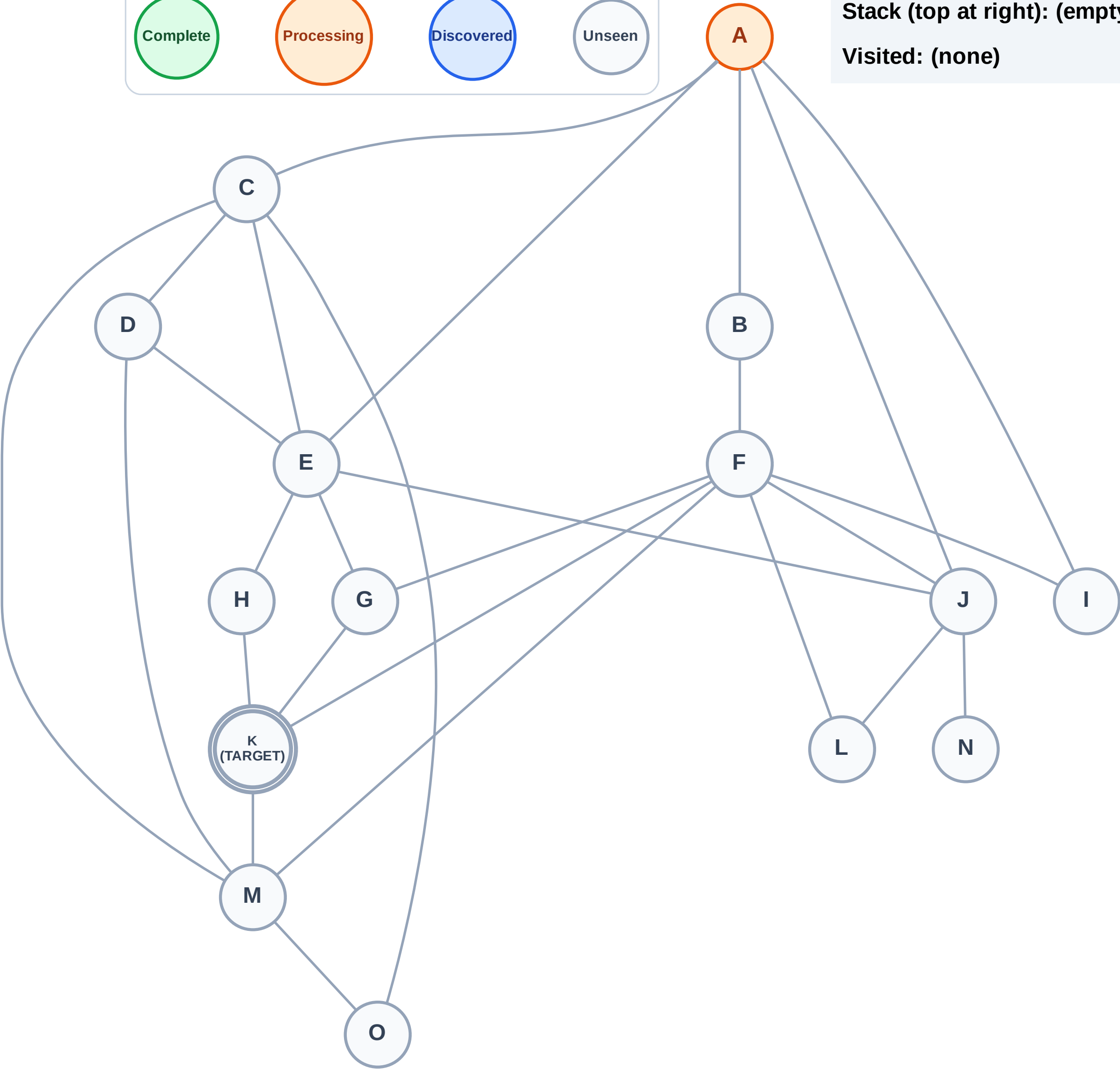
Complete

Processing

Discovered

Unseen

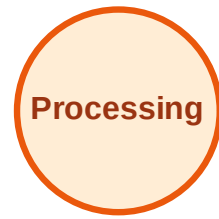
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

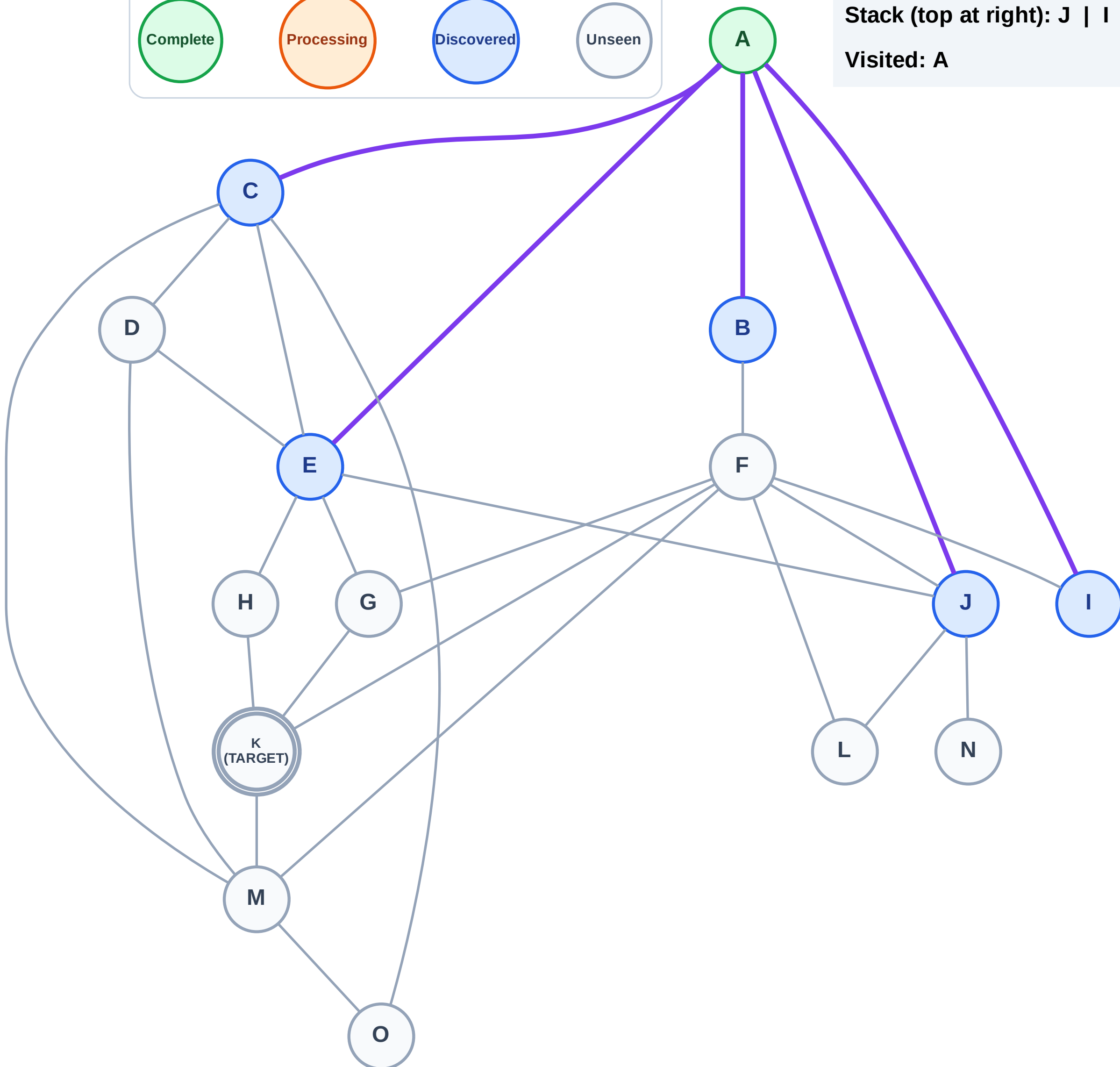
Step 3: Discover J, I, E, C, B from A

Legend



Stack (top at right): J | I | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

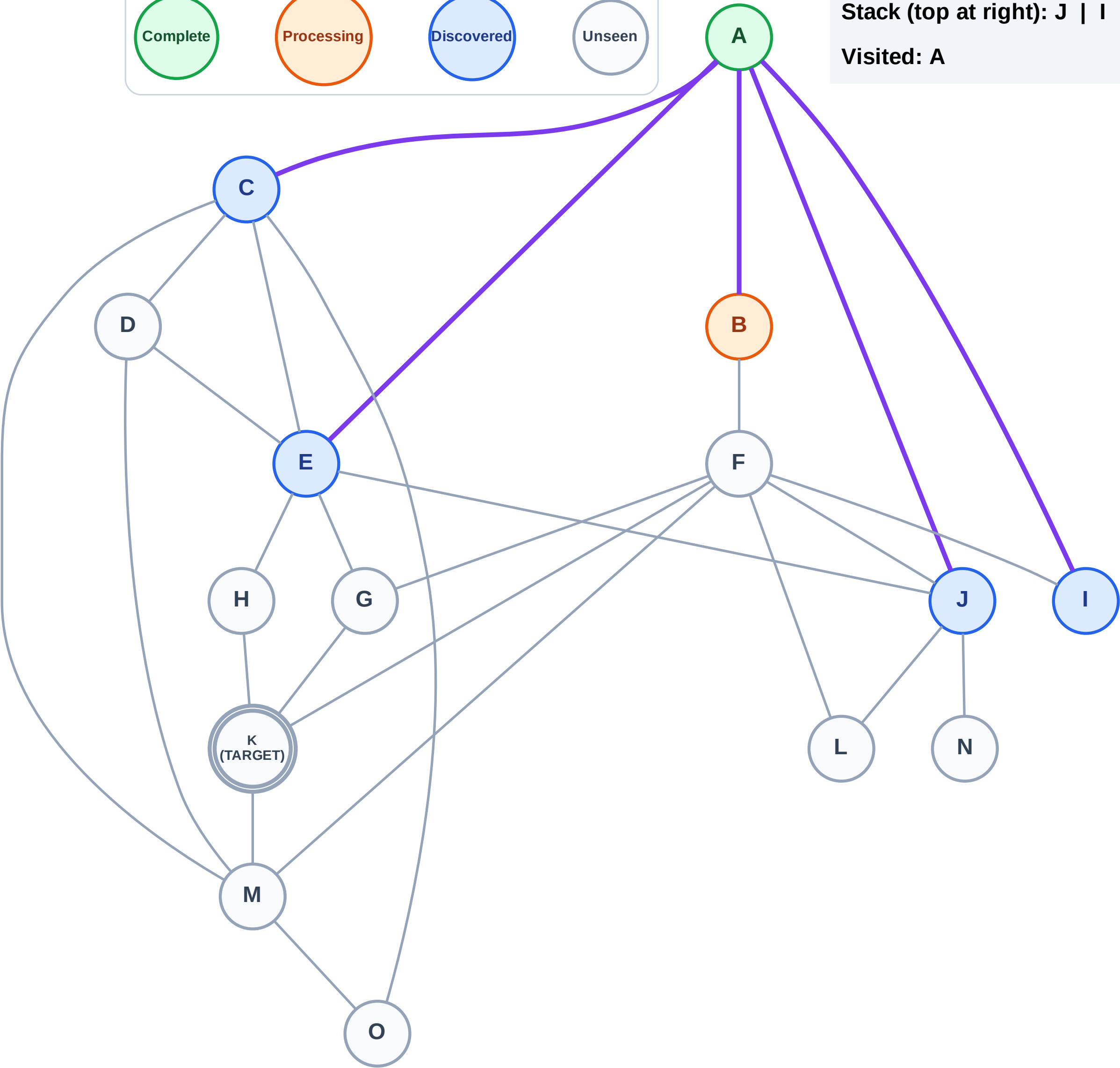
Processing

Discovered

Unseen

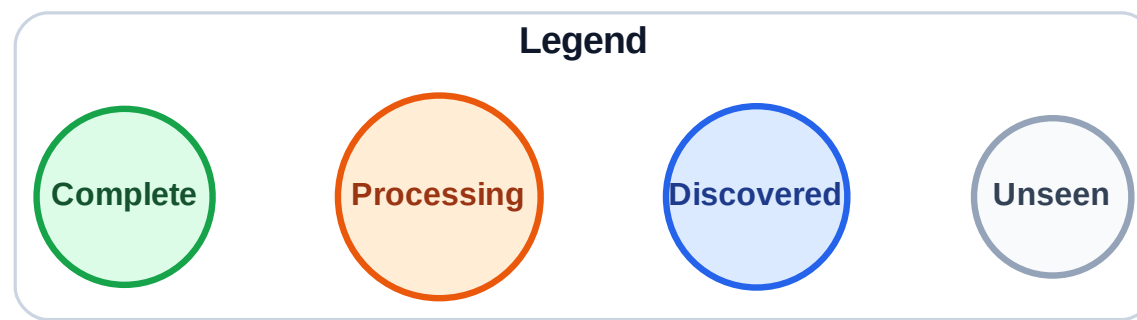
Stack (top at right): J | I | E | C

Visited: A



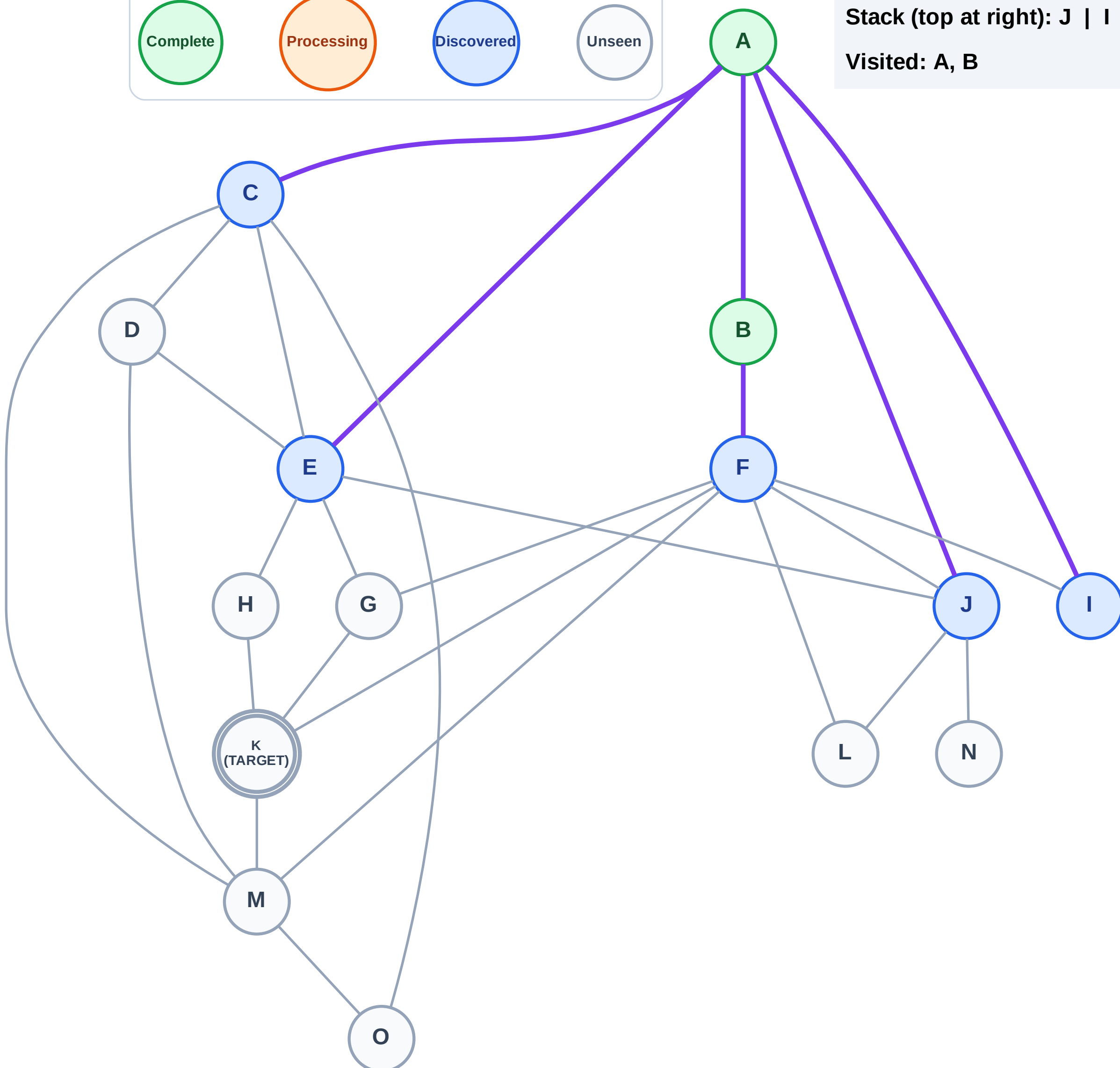
DFS Traversal

Step 5: Discover F from B



Stack (top at right): J | I | E | C | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

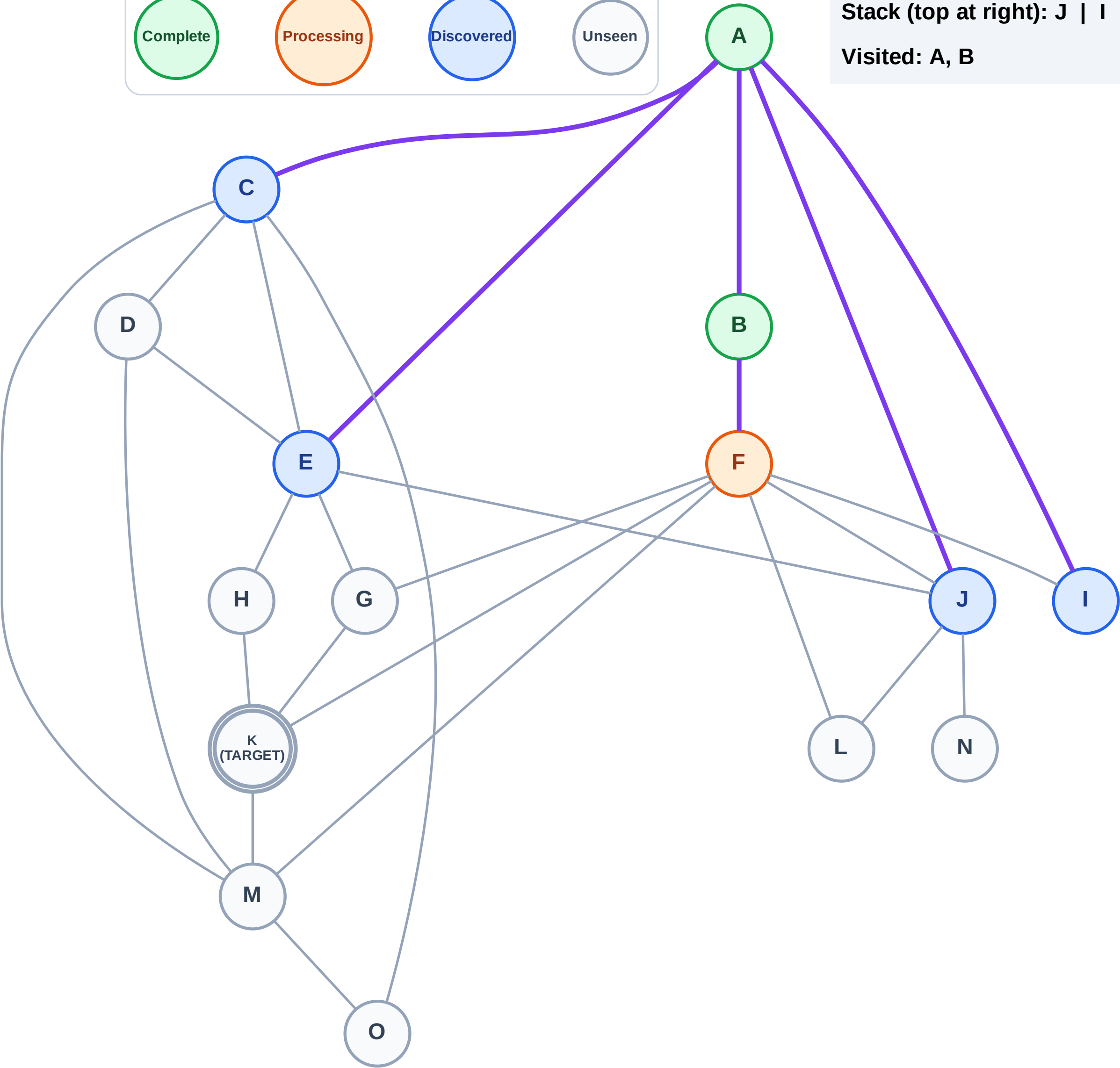
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | E | C
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | E | C | M | L

Visited: A, B, F, K

